

Object Oriented Programming (CSC166)

Lab Sheet 7

1. Write a function template Largest that returns the larger of two values of the same generic type. Use it to find the larger of two integers, two floats, and two characters.
2. Create a class template Pair that stores two values of a generic type through a parameterized constructor. Include functions to calculate and display their sum and difference. Instantiate the template with int and with float.
3. Design a template base class Item that stores an item code and a price of generic type T. Derive a regular class DiscountedItem from Item<double> that stores a discount percentage and calculates the final payable price. Display the item details and final price.
4. Write a program that reads the marks of a student and throws an exception if the marks are negative or greater than 100. Use try, throw, and catch to validate the input and display an appropriate error message.
5. Create a class Calculator with a member function divide(a, b) that performs division of two integers. Inside this function, throw an int exception if the divisor b is zero, and throw a string exception if either input value is negative. In main(), call divide() inside a try block with different sets of values, and use multiple catch blocks — one for int and one for string — to catch and display an appropriate message for each type of exception.